

FIELD VISIT PHOTOGRAPHS

Image.1

The prototype device is set up at the boarding edge during field testing at Dindoshi Bus Depot (BEST Route 326). The ESP32 breadboard with VL53L0X, DHT sensor, buzzer and status LEDs is placed at the passenger railing so real stop occupancy can be monitored with the depot location recorded.

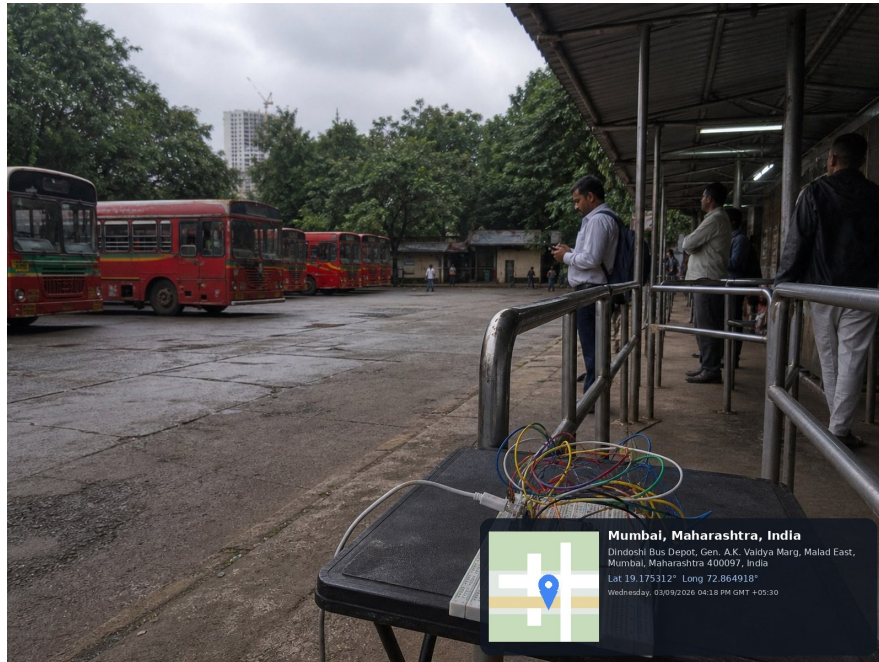


Image.1: Field Demonstration

Image.2

Close-up view of the enclosed working prototype used for sensor-based overcrowding monitoring: ESP32 Wi-Fi board, VL53L0X I2C ToF module (SDA/SCL), DHT temperature-humidity sensor, piezo buzzer, and red/green LED indicators with current-limiting resistors on the 3.3 V rail.

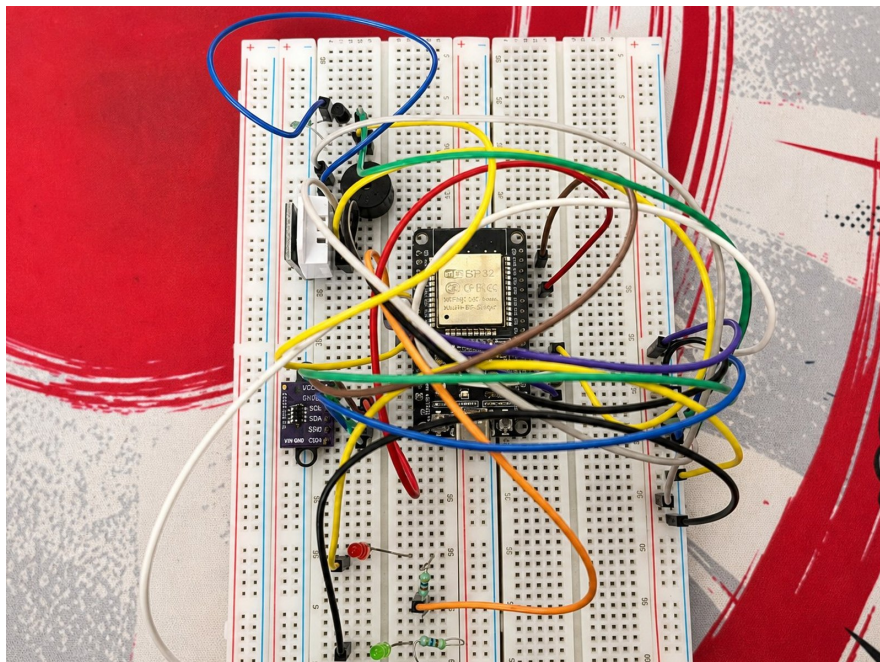


Image.2: Close-up Prototype Hardware

Image.3

A second laboratory close-up of the same breadboard after enhancement for report clarity. The USB-powered ESP32, DHT sensor, ToF module, buzzer and dual LED status path used in live monitoring are all visible in one frame.

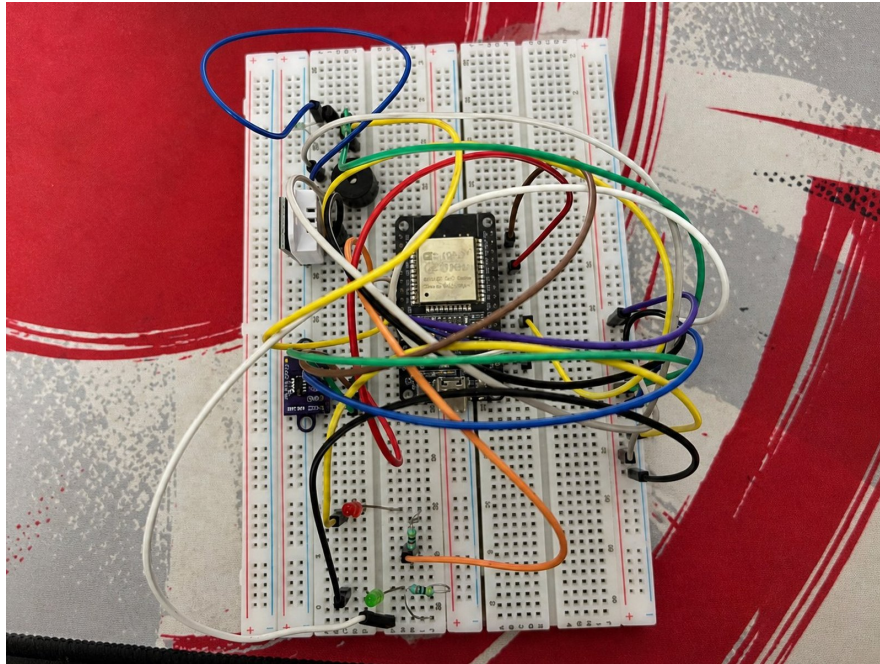


Image.3: Prototype Circuit (Enhanced Close-up)

Image.4

This image shows the Dindoshi stop shelter during the field visit. Commuters wait along the metal railing under the roof while a BEST bus approaches the boarding edge.



Image.4: Field Observation of the Bus Shelter

Image.5

Wide view of Dindoshi Bus Depot recorded during the visit. Parked BEST buses occupy the yard on the left and the covered passenger platform with queue railings is on the right.



Image.5: Dindoshi Bus Depot Yard

Image.6

Commuters queued under the Dindoshi Bus Station shelter. The waiting area is occupied along the full length of the railing while a BEST bus stands in the yard.



Image.6: Field Observation of an Overcrowded Bus Stop

The crowded platform is a real-world example of the occupancy condition that the IoT risk model is designed to flag for Route 326.